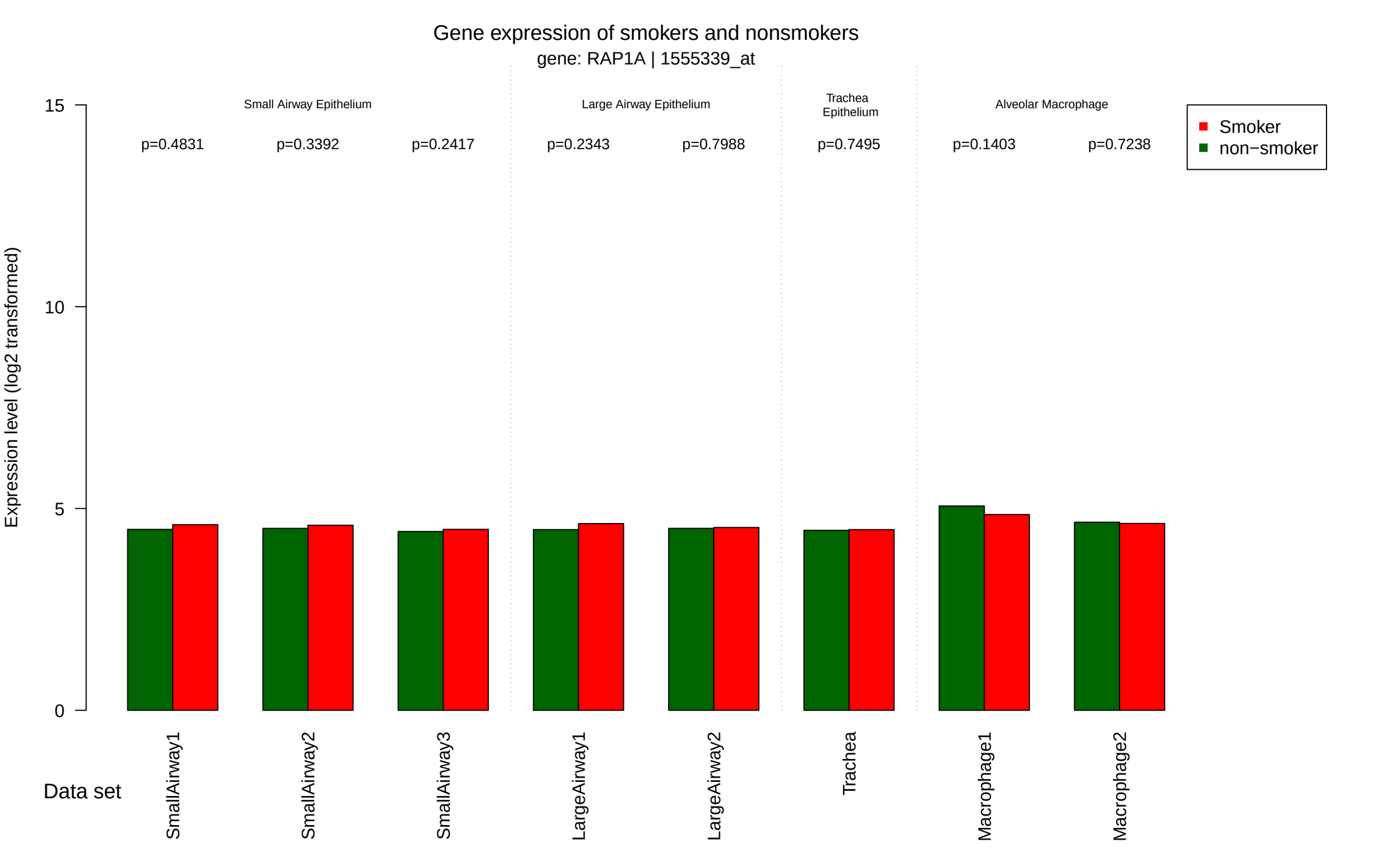
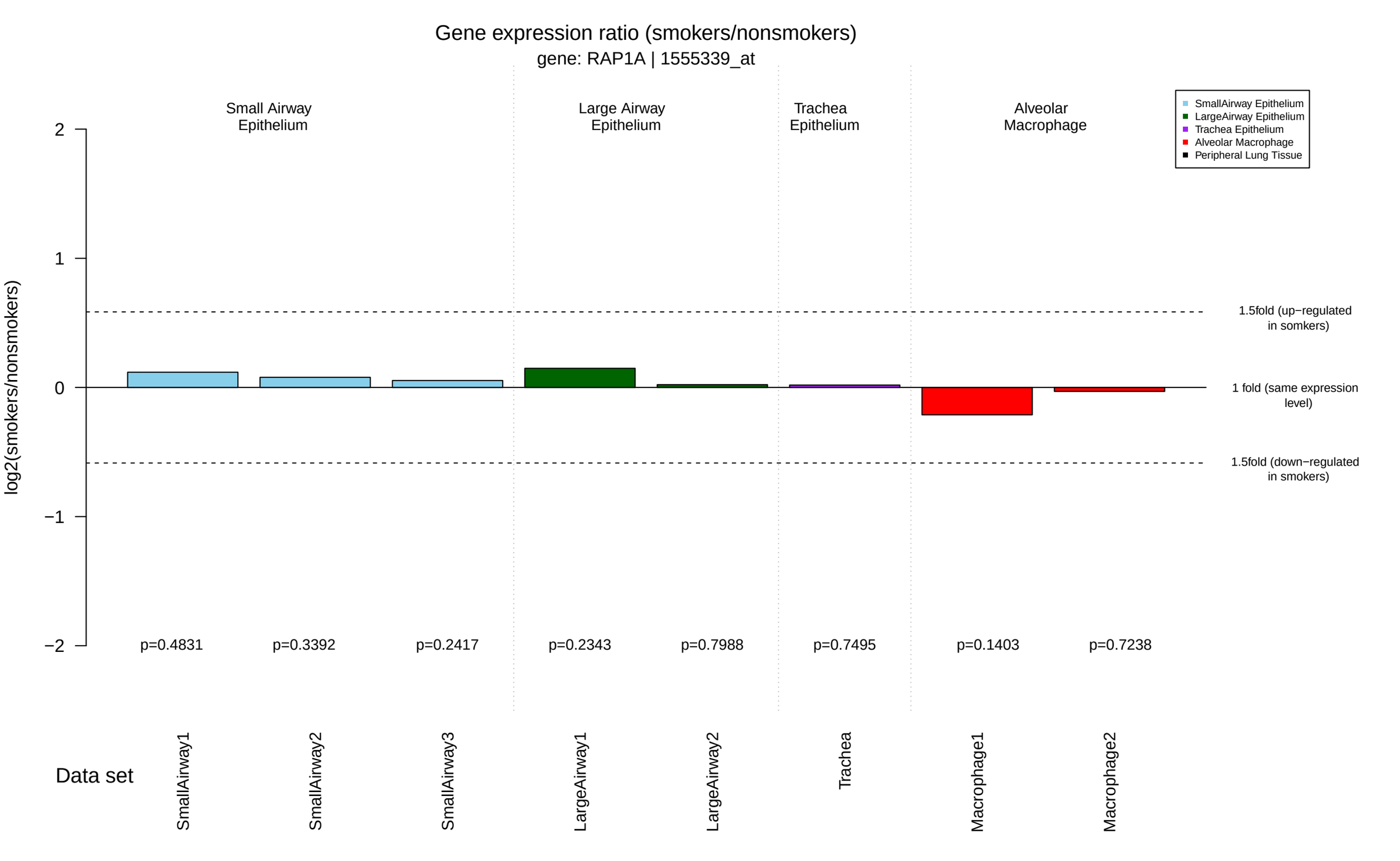
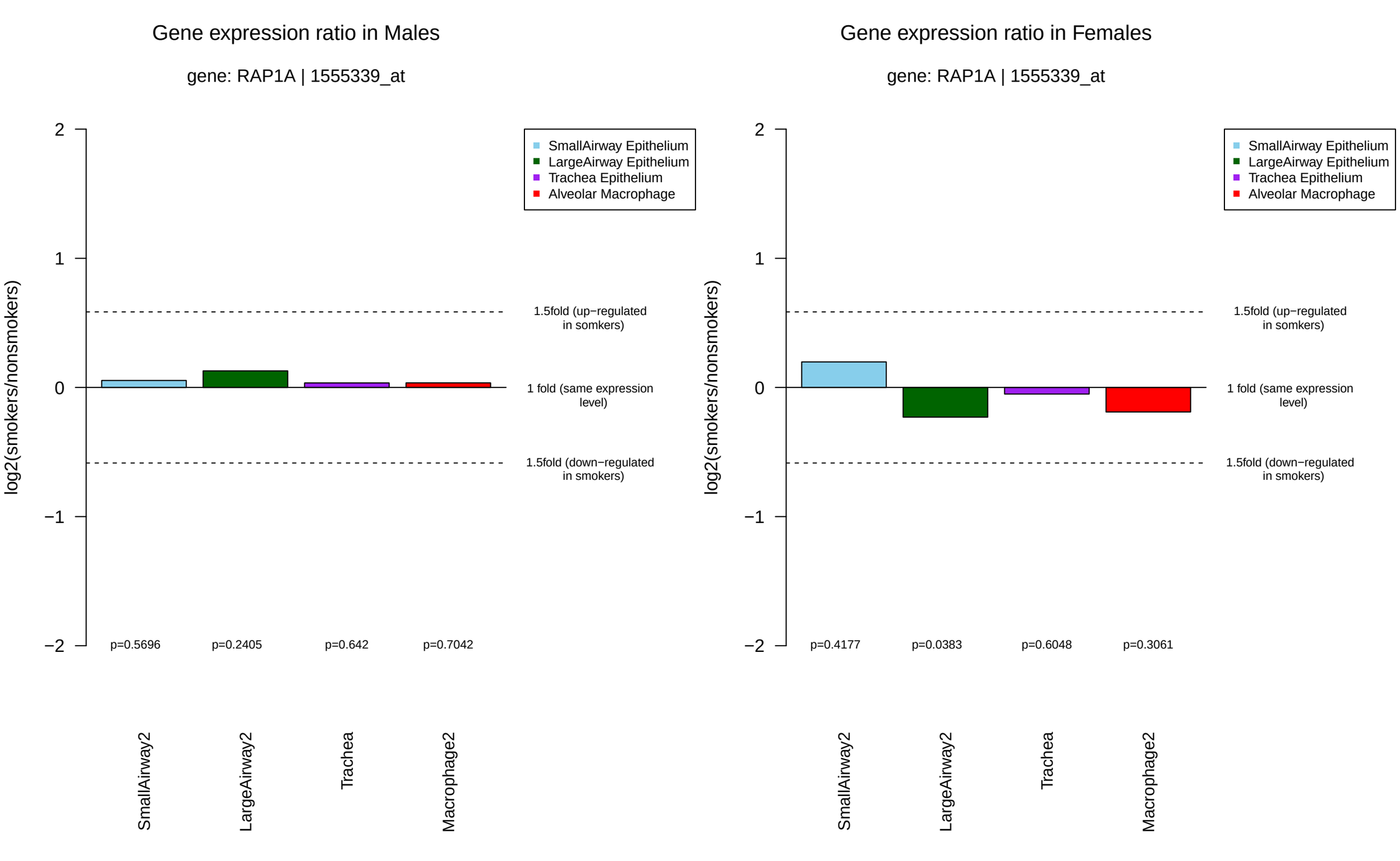
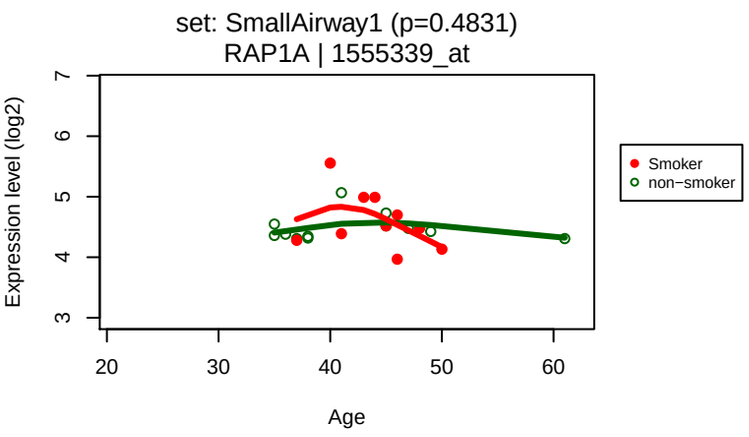
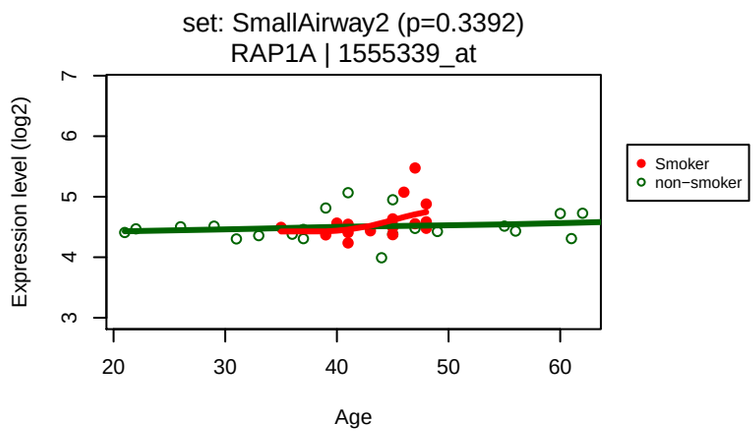
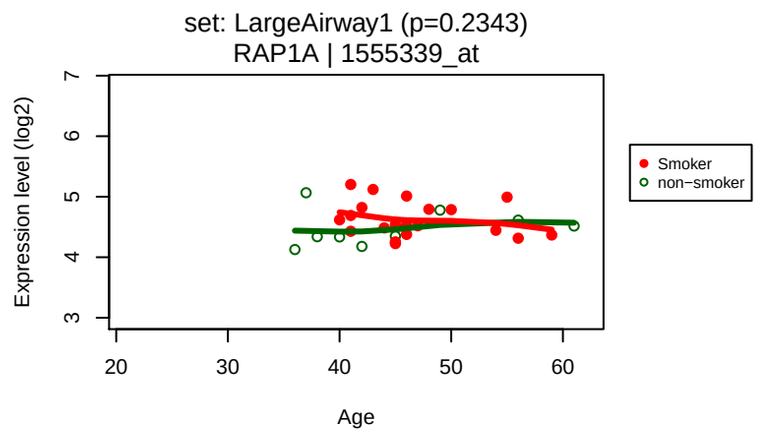

Gene expression of smokers and nonsmokers
gene: RAP1A | 1555339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4831
p=0.3392
p=0.2417
p=0.2343
p=0.7988
p=0.7495
p=0.1403
p=0.7238
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

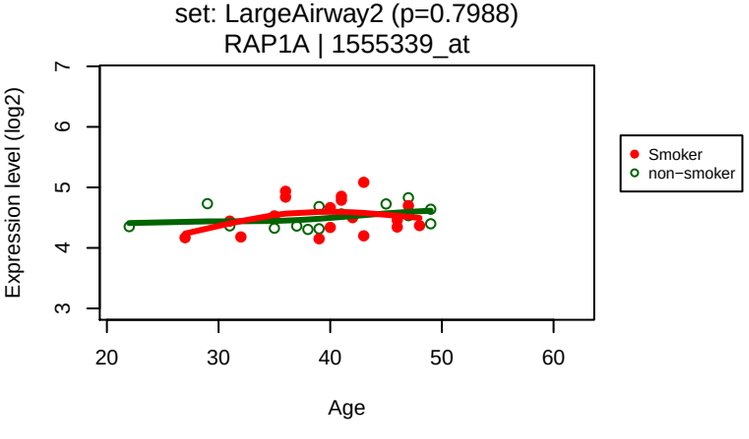

Gene expression ratio (smokers/nonsmokers)
gene: RAP1A | 1555339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4831
p=0.3392
p=0.2417
p=0.2343
p=0.7988
p=0.7495
p=0.1403
p=0.7238
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

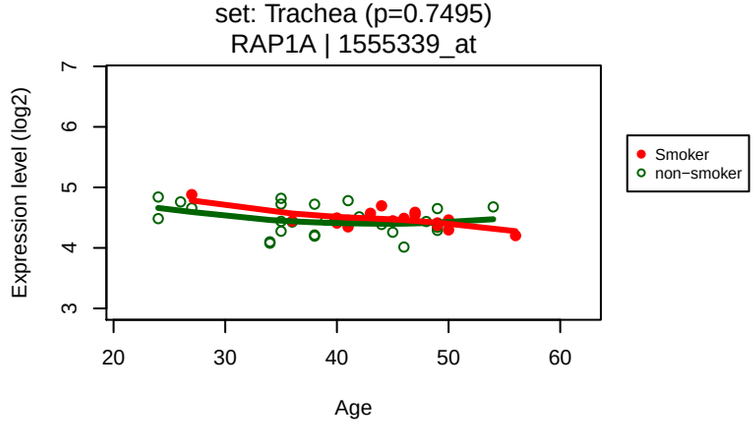

Gene expression ratio in Males
gene: RAP1A | 1555339_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5696
p=0.2405
p=0.642
p=0.7042
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP1A | 1555339_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4177
p=0.0383
p=0.6048
p=0.3061
SmallAirway2
LargeAirway2
Trachea
Macrophage2

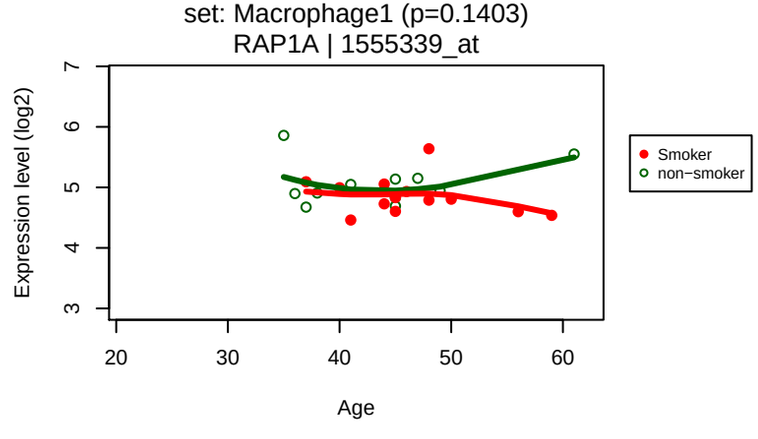

set: SmallAirway1 (p=0.4831)
RAP1A | 1555339_at
Expression level (log2)
Age
Smoker
non−smoker

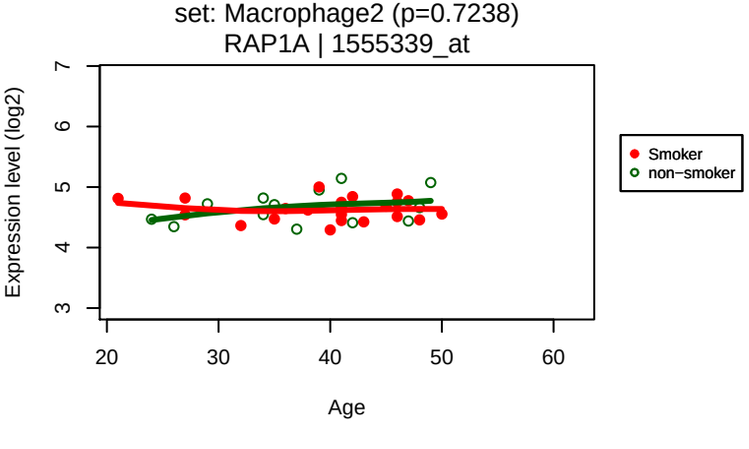

set: SmallAirway2 (p=0.3392)
RAP1A | 1555339_at
Expression level (log2)
Age
Smoker
non−smoker

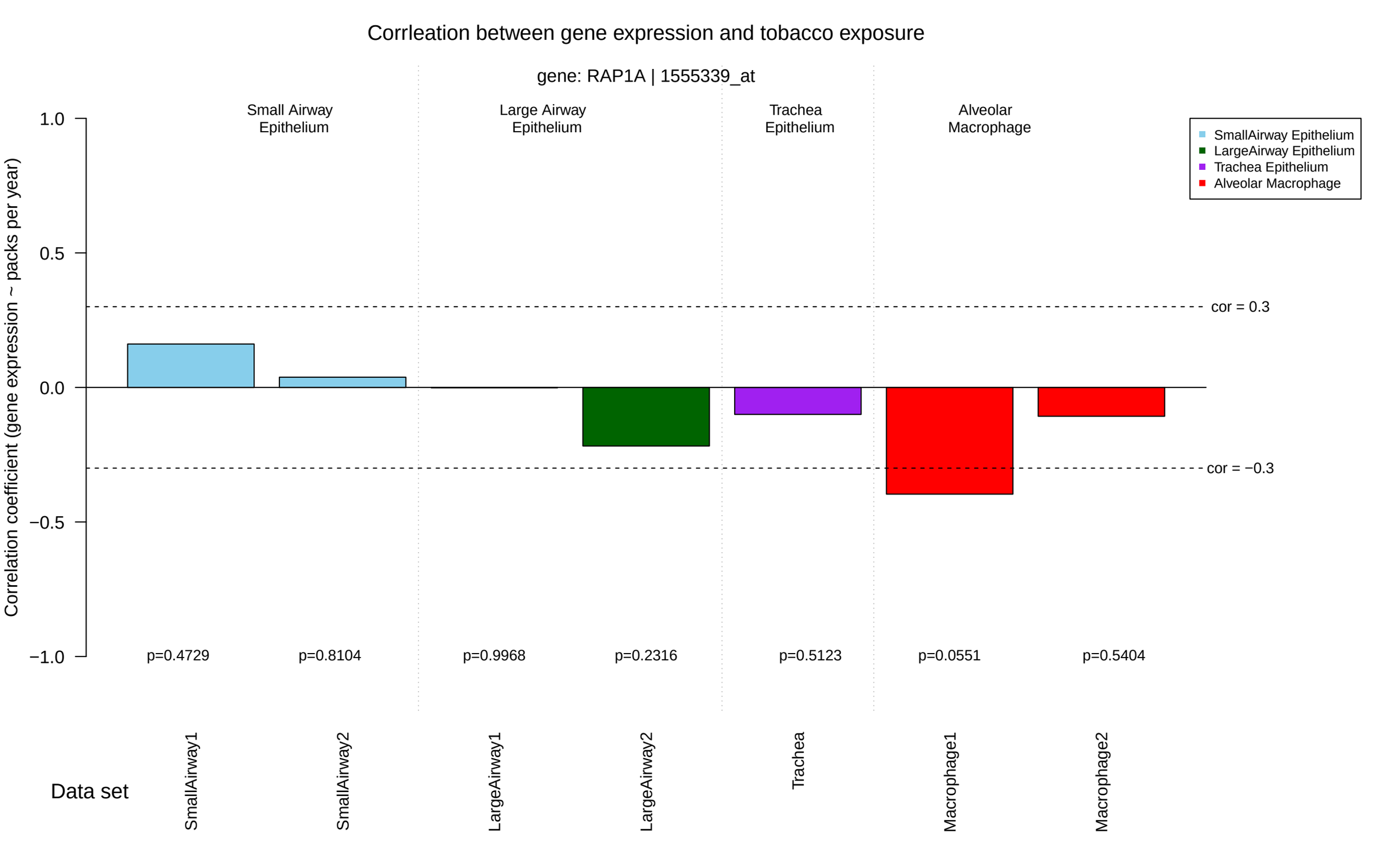

set: LargeAirway1 (p=0.2343)
RAP1A | 1555339_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7988)
RAP1A | 1555339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7495)
RAP1A | 1555339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1403)
RAP1A | 1555339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7238)
RAP1A | 1555339_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP1A | 1555339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4729
p=0.8104
p=0.9968
p=0.2316
p=0.5123
p=0.0551
p=0.5404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2